

Proofs are Programs

Tactics

Short Recap

Polymorphism

```
Inductive list (X:Type) : Type :=  
  | nil  
  | cons (x : X) (l : list X).
```

Higher-order functions

```
Fixpoint filter {X:Type}  
  (test: X->bool)  
  (l:list X) : list X :=  
  ...
```

Anonymous Functions

```
fun n => n * n
```

Functions that construct functions

```
Definition constfun {X: Type} (x: X)  
  : nat -> X :=  
  fun (k:nat) => x.
```

Applying Hypotheses

- General logical principle: if P and $P \rightarrow Q$ hold then also Q holds
- In other words: for showing Q if $P \rightarrow Q$ is known then it is sufficient to show P
- In particular: for showing P if P is known then there is nothing left to show (because P is equivalent to $True \rightarrow P$)
- Also: for showing $Q(c)$ if $\forall x. P(x) \rightarrow Q(x)$ is known then it is sufficient to show $P(c)$

Properties of Constructors

① Injectivity

Constructors need to be **injective**

- A function f is injective if $\forall xy. f(x) = f(y) \rightarrow x = y$

Properties of Constructors

② Disjointness

Constructors need to be **disjoint**

- Functions f and g are disjoint if $\forall xy. f(x) \neq g(y)$

Quiz 1

```
Inductive rgb : Type :=  
| red  
| green  
| blue.
```

```
Inductive color : Type :=  
| black  
| white  
| primary (p : rgb).
```

Suppose Coq's proof state looks like

```
x : rgb  
y : rgb  
H : primary x = primary y  
=====  
y = x
```



<https://etc.ch/hJM2>

and we apply the tactic `injection H as Hxy`. What will happen?

- 1) "No more subgoals."
- 2) The tactic fails.
- 3) Hypothesis `H` becomes `Hxy : x = y`.
- 4) None of the above.

Quiz 2

Suppose Coq's proof state looks like

```
x : bool
y : bool
H : negb x = negb y
=====
y = x
```

and we apply the tactic `injection H as Hxy`. What will happen? <https://etc.ch/hJM2>

- 1) "No more subgoals."
- 2) The tactic fails.
- 3) Hypothesis `H` becomes `Hxy : x = y`.
- 4) None of the above.



Quiz 3

Now suppose Coq's proof state looks like

```
x : nat
y : nat
H : x + 1 = y + 1
=====
y = x
```

and we apply the tactic `injection H as Hxy`. What will happen?

- 1) "No more subgoals."
- 2) The tactic fails.
- 3) Hypothesis `H` becomes `Hxy : x = y`.
- 4) None of the above.



<https://etc.ch/hJM2>

Quiz 4

Finally, suppose Coq's proof state looks like

```
x : nat
y : nat
H : 1 + x = 1 + y
=====
y = x
```

and we apply the tactic `injection H as Hxy`. What will happen?

- 1) "No more subgoals."
- 2) The tactic fails.
- 3) Hypothesis `H` becomes `Hxy : x = y`.
- 4) None of the above.



<https://etc.ch/hJM2>

Properties of Constructors

③ Functionality

Constructors are functions (and, hence **functional**)

- $\forall xy. x = y \rightarrow f(x) = f(y)$

Modifying Hypotheses

Specialising Hypotheses

Strengthening the Induction Hypothesis

$$\forall nm. \text{double } n = \text{double } m \rightarrow n = m$$

```
Fixpoint double (n:nat) :=  
  match n with  
  | 0 => 0  
  | S n' => S (S (double n'))  
end.
```

Unfolding Definitions

Summary

Applying lemmas and hypotheses

`apply`

`apply ... with (x := ...)`

Exploiting properties of constructors

`injection ... as ...`

`discriminate`

Generalising inductive hypotheses

`generalize dependent`

Other tactics

`symmetry`

`transitivity`

`unfold`